Marco Del Fava

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Summary

Recently admitted to the UC Davis M.S. in Biomedical Engineering (Fall 2026) and active neuroscience researcher working at the intersection of human intracranial electrophysiology, single-unit digital signal processing, and neurotechnology tooling. Experience includes spike sorting with Kilosort4, O-Sort, and SpikeInterface; building GPU-accelerated pipelines on AWS; and implementing Dark Horse Neuro / Neuralynx acquisition workflows for replay, compression-aware MED recording, and large-scale neural data handling. I look forward to a career in neural signal processing, neural acquisition systems, and BCI-related infrastructure.

Neuroscience Projects

Dark Horse Neuro / Neuralynx acquisition pipeline | 2026

- Built a Python orchestration layer to launch DHN-AQ headlessly over SSH, generate RC/CS session files, and coordinate Pegasus / NetCom replay workflows for human intracranial recordings.
- Implemented an NRD replay bridge that re-broadcasts ATLAS-format UDP packets so archived Neuralynx sessions can be ingested by DHN-AQ as if hardware were live.
- Automated compression-aware MED recording setup for 512-channel macro / micro configurations, including decimation, lossless/lossy options, and pre-flight network diagnostics.
- Links: GitHub · Docs **spike_discrim single-unit DSP workbench** | 2026
- Benchmarked low-latency waveform representations for real-time single-unit discrimination using curated OSort snippets, synthetic data, AUC, balanced accuracy, mutual information, and KNN purity.

- Built a Numba-accelerated WeightBank population code to score spike-likeness and project waveforms into afferent activation space for clustering and unit-separation analysis.

- Links: GitHub **SNN Agent real-time spike sorting** | 2026

- Developed a streaming spiking neural network with bandpass filtering, Kalman noise gating, competitive STDP, and browser-based monitoring for live spike detection and sorting.

- Evaluated 20 kHz real-time pipeline behavior across synthetic, .ncs, UDP, and LSL inputs with online decoding for closed-loop neurotech use cases.

- **Links: GitHub**

Experience

Operations Specialist — First Step Communities | Sacramento, CA | Jul 2021 – Present

- Analyzed program and partner datasets using Python, SPSS, and Excel to identify predictors of successful exits and service effectiveness.
- Coordinated data collection from clinical partners including UC Davis Health and Kaiser Permanente to support integrated care outcome analyses.
- Identified an association between on-site clinical engagement and permanent housing outcomes; supported leadership in scaling integrated clinical services across sites.
- Developed 40+ standardized data collection forms and automated reporting workflows using Microsoft Power Automate.

Analyst — Eurofins Test America | West Sacramento, CA | Oct 2019 – Jun 2021

- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.
- Performed solvent extraction of organic compounds from soil and water samples in a high-throughput production setting.
- Coordinated daily scheduling and backlog management to meet turnaround targets across teams.

Research & Volunteer Experience

Research Volunteer — UC Davis Health — Z-Lab (Dr. Jie Zheng) | Sacramento, CA | Jul 2025 – Present

- Build a GPU-accelerated spike sorting pipeline for dense microelectrode arrays using Kilosort4 on AWS EC2 instances to support reproducible, high-throughput reprocessing of human intracranial recordings.

- Sort single-unit activity from human Phase II epilepsy microwire recordings using MATLAB (O-Sort) and Python (SpikeInterface), with an emphasis on preserving unit-specific waveform structure for downstream analysis.
- Curated a dataset of approximately 2 million hand-labeled waveforms for spike-vs-noise modeling and benchmarking.
- Trained neural network models to classify putative single units versus noise, improving manual curation efficiency for human recordings.

Research Volunteer — UC Davis Health — Neuroscience Unit E5 | Sacramento, CA | Dec 2022 – Jul 2023

- Prepared headboxes and EEG application kits for the Epilepsy Monitoring Unit.
- Shadowed EEG technologists and observed clinical neurophysiology procedures for patients with epilepsy and traumatic brain injury.

Research Intern — California Home Builders Association | Sacramento, CA | Jul 2023 – Feb 2024

- Designed a cost-estimation model for emergency shelter villages with on-site medical facilities to inform California housing policy.
- Synthesized findings into policy recommendations to support emergency shelter development legislation.

Education

M.S. Biomedical Engineering | University of California, Davis | Beginning Fall 2026 Recently admitted; incoming first-year master's student

Python for Data Science, Professional Concentration (GPA: 4.0) | UC Davis Continuing and Professional Education | 2025

M.S. Psychology (GPA: 3.7) | Arizona State University | 2023

B.S. Neuroscience, Physiology, and Behavior (GPA: 3.1) | University of California, Davis | 2017

Relevant Coursework

- Neurobiology of Behavior — UC Davis
- Physiology of Psychology — Arizona State University
- Quantitative Analysis — Arizona State University
- Research Methods — Arizona State University

Skills

Neural Data & Signal Processing: Single-unit DSP and waveform feature extraction, Spike sorting (Kilosort4, O-Sort, SpikeInterface), Neural data processing (filtering, PCA, template matching, clustering), Intracranial electrophysiology (microwire, LFP, iEEG)

Neural Acquisition & Tooling: Neuralynx Pegasus / ATLAS / NetCom workflows, Dark Horse Neuro DHN-AQ RC/CS configuration, NRD replay, UDP relay, and MED archive generation, Linux, SSH automation, xvfb, and acquisition diagnostics

Machine Learning & Data Science: Python (NumPy, Pandas, scikit-learn, PyTorch, Numba), Statistical analysis (SPSS), SQL and database engineering (PostgreSQL, PGVector), Cloud GPU computing (AWS EC2, CUDA, Docker) and MATLAB workflows

General Technical: Git and version control, Linux and shell scripting, Web interfaces (Streamlit), Microsoft 365 administration

Data Systems & Reporting: Dashboards and stakeholder reporting (Looker), Workflow automation (Microsoft 365, Power Automate), Survey design and focus group facilitation, Equity-informed outcomes measurement

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Certifications

- **Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries** — UC Davis Continuing and Professional Education (2025-09-18) — View badge
- CITI — Biomedical Research
- CITI — Essentials of Research Administration
- CITI — GCP Social and Behavioral Research Best Practices